

Identity. This is simply DDIM with x -prediction network.

$$\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) = (\alpha_s - \frac{\sigma_s}{\sigma_t}\alpha_t)\mathbf{G}_\theta\left(\frac{\mathbf{x}_t}{\sigma_d\sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t)\right) + \frac{\sigma_s}{\sigma_t}\mathbf{x}_t$$

Simple-EDM.

$$\begin{aligned}\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) &= (\alpha_s - \frac{\sigma_s}{\sigma_t}\alpha_t)\left[\frac{\alpha_t}{\alpha_t^2 + \sigma_t^2}\mathbf{x}_t - \frac{\sigma_t}{\sqrt{\alpha_t^2 + \sigma_t^2}}\sigma_d\mathbf{G}_\theta\left(\frac{\mathbf{x}_t}{\sigma_d\sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t)\right)\right] + \frac{\sigma_s}{\sigma_t}\mathbf{x}_t \\ &= \frac{\alpha_s\alpha_t + \sigma_s\sigma_t}{\alpha_t^2 + \sigma_t^2}\mathbf{x}_t - \frac{\alpha_s\sigma_t - \sigma_s\alpha_t}{\sqrt{\alpha_t^2 + \sigma_t^2}}\sigma_d\mathbf{G}_\theta\left(\frac{\mathbf{x}_t}{\sigma_d\sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t)\right)\end{aligned}$$

When noise schedule is cosine, $\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) = \cos(\frac{1}{2}\pi(t-s))\mathbf{x}_t - \sin(\frac{1}{2}\pi(t-s))\sigma_d\mathbf{G}_\theta\left(\frac{\mathbf{x}_t}{\sigma_d\sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t)\right)$. And similar to [Lu & Song \(2024\)](#), we can show that predicting $\mathbf{x}_s = \text{DDIM}(\mathbf{x}_t, \mathbf{x}, s, t)$ with $\mathcal{L}_2$ loss is equivalent to v -prediction with cosine schedule.

$$\begin{aligned}w(s, t) &\left\|\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \left((\alpha_s - \frac{\sigma_s}{\sigma_t}\alpha_t)\mathbf{x} + \frac{\sigma_s}{\sigma_t}\mathbf{x}_t\right)\right\|^2 \\ &= w(s, t) \left\|\frac{\alpha_s\alpha_t + \sigma_s\sigma_t}{\alpha_t^2 + \sigma_t^2}\mathbf{x}_t - \frac{\alpha_s\sigma_t - \sigma_s\alpha_t}{\sqrt{\alpha_t^2 + \sigma_t^2}}\sigma_d\mathbf{G}_\theta\left(\frac{\mathbf{x}_t}{\sigma_d\sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t)\right) - \left((\alpha_s - \frac{\sigma_s}{\sigma_t}\alpha_t)\mathbf{x} + \frac{\sigma_s}{\sigma_t}\mathbf{x}_t\right)\right\|^2 \\ &= \tilde{w}(s, t) \left\|\sigma_d\mathbf{G}_\theta\left(\frac{\mathbf{x}_t}{\sigma_d\sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t)\right) - \underbrace{\left(-\frac{\sqrt{\alpha_t^2 + \sigma_t^2}}{\alpha_s\sigma_t - \sigma_s\alpha_t}\right)\left((\alpha_s - \frac{\sigma_s}{\sigma_t}\alpha_t)\mathbf{x} + \frac{\sigma_s}{\sigma_t}\mathbf{x}_t - \frac{\alpha_s\alpha_t + \sigma_s\sigma_t}{\alpha_t^2 + \sigma_t^2}\mathbf{x}_t\right)}_{\mathbf{G}_{\text{target}}}\right\|^2\end{aligned}$$

where

$$\begin{aligned}\mathbf{G}_{\text{target}} &= \left(-\frac{\sqrt{\alpha_t^2 + \sigma_t^2}}{\alpha_s\sigma_t - \sigma_s\alpha_t}\right)\left((\alpha_s - \frac{\sigma_s}{\sigma_t}\alpha_t)\mathbf{x} + \frac{\sigma_s}{\sigma_t}\mathbf{x}_t - \frac{\alpha_s\alpha_t + \sigma_s\sigma_t}{\alpha_t^2 + \sigma_t^2}\mathbf{x}_t\right) \\ &= \left(-\frac{\sqrt{\alpha_t^2 + \sigma_t^2}}{\alpha_s\sigma_t - \sigma_s\alpha_t}\right)\left((\alpha_s - \frac{\sigma_s}{\sigma_t}\alpha_t)\mathbf{x} + \frac{\sigma_s\alpha_t^2 + \cancel{\sigma_s\sigma_t^2} - \alpha_s\alpha_t\sigma_t - \cancel{\sigma_s\sigma_t^2}}{\sigma_t(\alpha_t^2 + \sigma_t^2)}(\alpha_t\mathbf{x} + \sigma_t\epsilon)\right) \\ &= \left(-\frac{\sqrt{\alpha_t^2 + \sigma_t^2}}{\alpha_s\sigma_t - \sigma_s\alpha_t}\right)\left(\frac{(\alpha_s\sigma_t - \sigma_s\alpha_t)(\alpha_t^2 + \sigma_t^2) + \sigma_s\alpha_t^3 - \alpha_s\alpha_t^2\sigma_t}{\sigma_t(\alpha_t^2 + \sigma_t^2)}\mathbf{x} + \frac{\sigma_s\alpha_t^2 - \alpha_s\alpha_t\sigma_t}{\alpha_t^2 + \sigma_t^2}\epsilon\right) \\ &= \left(-\frac{\sqrt{\alpha_t^2 + \sigma_t^2}}{\alpha_s\sigma_t - \sigma_s\alpha_t}\right)\left(\frac{(\alpha_s\sigma_t - \sigma_s\alpha_t)(\alpha_t^2 + \sigma_t^2) - (\alpha_s\sigma_t - \sigma_s\alpha_t)\alpha_t^2}{\alpha_t^2 + \sigma_t^2}\mathbf{x} - \frac{(\alpha_s\sigma_t - \sigma_s\alpha_t)\alpha_t}{\sigma_t(\alpha_t^2 + \sigma_t^2)}\epsilon\right) \\ &= \frac{\alpha_t\epsilon - \sigma_t\mathbf{x}}{\sqrt{\alpha_t^2 + \sigma_t^2}}\end{aligned}$$

This reduces to v -target if cosine schedule is used, and it deviates from v -target if FM schedule is used instead.

Euler-FM. We assume OT-FM schedule.

$$\begin{aligned}
 \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) &= \left((1-s) - \frac{s}{t}(1-t) \right) \left[\mathbf{x}_t - t\sigma_d \mathbf{G}_\theta \left(\frac{\mathbf{x}_t}{\sigma_d \sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t) \right) \right] + \frac{s}{t} \mathbf{x}_t \\
 &= \left((1 - \frac{s}{t}) \right) \left[\mathbf{x}_t - t\sigma_d \mathbf{G}_\theta \left(\frac{\mathbf{x}_t}{\sigma_d \sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t) \right) \right] + \frac{s}{t} \mathbf{x}_t \\
 &= \mathbf{x}_t - (t-s)\sigma_d \mathbf{G}_\theta \left(\frac{\mathbf{x}_t}{\sigma_d \sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t) \right)
 \end{aligned}$$

This results in Euler ODE from $\mathbf{x}_t$ to $\mathbf{x}_s$. We also show that the network output reduces to v -prediction if matched with $\mathbf{x}_s = \text{DDIM}(\mathbf{x}_t, \mathbf{x}, s, t)$. To see this,

$$\begin{aligned}
 w(s, t) &\left\| \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \left((1-s) - \frac{s}{t}(1-t) \right) \mathbf{x} + \frac{s}{t} \mathbf{x}_t \right\|^2 \\
 &= w(s, t) \left\| \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \left((1 - \frac{s}{t}) \mathbf{x} + \frac{s}{t} \mathbf{x}_t \right) \right\|^2 \\
 &= w(s, t) \left\| \mathbf{x}_t - (t-s)\sigma_d \mathbf{G}_\theta \left(\frac{\mathbf{x}_t}{\sigma_d \sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t) \right) - \left((1 - \frac{s}{t}) \mathbf{x} + \frac{s}{t} \mathbf{x}_t \right) \right\|^2 \\
 &= \tilde{w}(s, t) \left\| \sigma_d \mathbf{G}_\theta \left(\frac{\mathbf{x}_t}{\sigma_d \sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t) \right) - \left(-\frac{1}{(t-s)} \right) \left((1 - \frac{s}{t}) \mathbf{x} + (\frac{s}{t} - 1) \mathbf{x}_t \right) \right\|^2 \\
 &= \tilde{w}(s, t) \left\| \sigma_d \mathbf{G}_\theta \left(\frac{\mathbf{x}_t}{\sigma_d \sqrt{\alpha_t^2 + \sigma_t^2}}, c_{\text{noise}}(s), c_{\text{noise}}(t) \right) - \underbrace{\left(-\frac{1}{(t-s)} \right) \left((1 - \frac{s}{t}) \mathbf{x} + (\frac{s}{t} - 1) \mathbf{x}_t \right)}_{\mathbf{G}_{\text{target}}} \right\|^2
 \end{aligned}$$

where

$$\begin{aligned}
 \mathbf{G}_{\text{target}} &= \left(-\frac{1}{(t-s)} \right) \left((1 - \frac{s}{t}) \mathbf{x} + (\frac{s}{t} - 1) ((1-t) \mathbf{x} + t\epsilon) \right) \\
 &= \left(-\frac{1}{(t-s)} \right) \left((1 - \frac{s}{t}) \mathbf{x} + (\frac{s}{t} - s - 1 + t) \mathbf{x} + (s-t) \epsilon \right) \\
 &= \left(-\frac{1}{(t-s)} \right) ((t-s) \mathbf{x} + (s-t) \epsilon) \\
 &= \epsilon - \mathbf{x}
 \end{aligned}$$

which is v -target under OT-FM schedule. This parameterization naturally allows zero-SNR sampling and satisfies boundary condition at $s = 0$, similar to Simple-EDM above. This is not true for Identity parametrization using $\mathbf{G}_\theta$ as it satisfies boundary condition only at $s > 0$.

C.6. Mapping Function $r(s, t)$

We discuss below the concrete choices for $r(s, t)$. We use a constant decrement $\epsilon > 0$ in different spaces.

Constant decrement in $\eta(t) := \eta_t = \sigma_t / \alpha_t$. This is the choice that we find to work better than other choices in practice. First, let its inverse be $\eta^{-1}(\cdot)$,

$$r(s, t) = \max(s, \eta^{-1}(\eta(t) - \epsilon))$$

We choose $\epsilon = (\eta_{\max} - \eta_{\min}) / 2^k$ for some k . We generally choose $\eta_{\max} \approx 160$ and $\eta_{\min} \approx 0$. We find $k = \{10, \dots, 15\}$ works well enough depending on datasets.

Constant decrement in t .

$$r(s, t) = \max(s, t - \epsilon)$$

We choose $\epsilon = (T - \epsilon)/2^k$.

Constant decrement in $\lambda(t) := \log\text{-SNR}_t = 2\log(\alpha_t/\sigma_t)$.

Let its inverse be $\lambda^{-1}(\cdot)$, then

$$r(s, t) = \max(s, \lambda^{-1}(\lambda(t) - \epsilon))$$

We choose $\epsilon = (\lambda_{\max} - \lambda_{\min})/2^k$. This choice comes close to the first choice, but we refrain from this because $r(s, t)$ becomes close to t both when $t \approx 0$ and $t \approx 1$ instead of just $t \approx 1$. This gives more chances for training instability than the first choice.

Constant increment in $1/\eta(t)$.

$$r(s, t) = \max\left(s, \eta^{-1}\left(\frac{1}{1/\eta(t) + \epsilon}\right)\right)$$

We choose $\epsilon = (1/\eta(t)_{\min} - 1/\eta(t)_{\max})/2^k$.

C.7. Time Distribution $p(s, t)$

In all cases we choose $p(t) = \mathcal{U}(\epsilon, T)$ and $p(s|t) = \mathcal{U}(\epsilon, t)$ for some $\epsilon \geq 0$ and $T \leq 1$. The decision for time distribution is coupled with $r(s, t)$. We list the constraints on $p(s, t)$ for each $r(s, t)$ choice below.

Constant decrement in $\eta(t)$. We need to choose $T < 1$ because, for example, assuming OT-FM schedule, $\eta_t = t/(1 - t)$, one can observe that constant decrement in η_t when $t \approx 1$ results in $r(s, t)$ that is too close to t due to η_t 's exploding gradient around 1. We need to define $T < 1$ such that $r(s, T)$ is not too close to T for s reasonably far away. With $\eta_{\max} \approx 160$, we can choose $T = 0.994$ for OT-FM and $T = 0.996$ for VP-diffusion.

Constant decrement in t . No constraints needed. $T = 1$, $\epsilon = 0$.

Constant decrement in λ_t . One can similarly observe exploding gradient causing $r(s, t)$ to be too close to t at both $t \approx 0$ and $t \approx 1$, so we can choose $\epsilon > 0$, e.g. 0.001, in addition to choosing $T = 0.994$ for OT-FM and $T = 0.996$ for VP-diffusion.

Constant increment in $1/\eta_t$. This experience exploding gradient for $t \approx 0$, so we require $\epsilon > 0$, e.g. 0.005. And $T = 1$.

C.8. Kernel Function

For our Laplace kernel $k(x, y) = \exp(-\tilde{w}(s, t) \max(\|x - y\|_2, \epsilon)/D)$, we let $\epsilon > 0$ be a reasonably small constant, e.g. 10^{-8} . Looking at its gradient w.r.t. x ,

$$\nabla_x e^{-\tilde{w}(s, t) \max(\|x - y\|_2, \epsilon)/D} = \begin{cases} -\frac{\tilde{w}(s, t)}{D} e^{-\tilde{w}(s, t) \|x - y\|_2/D} \frac{x - y}{\|x - y\|_2}, & \text{if } \|x - y\|_2 > \epsilon \\ 0, & \text{otherwise} \end{cases}$$

one can notice that the gradient is self-normalized to be a unit vector, which is helpful in practice. In comparison, the gradient of RBF kernel of the form $k(x, y) = \exp(-\frac{1}{2} \tilde{w}(s, t) \|x - y\|^2/D)$,

$$\nabla_x e^{-\tilde{w}(s, t) \|x - y\|^2/D} = -\frac{\tilde{w}(s, t)}{D} e^{-\tilde{w}(s, t) \frac{1}{2} \|x - y\|^2/D} (x - y) \quad (68)$$

whose magnitude can vary a lot depending on how far x is from y .

For $\tilde{w}(s, t)$, we find it helpful to write out the $\mathcal{L}_2$ loss between the arguments. For simplicity, we denote $\hat{\mathbf{G}}_\theta(\mathbf{x}_t, s, t) =$